Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2024-25

Class:	SE	Semester:	III
Course Code:	CSL304	Course Name:	OBJECT ORIENTED PROGRAMMING IN JAVA

Name of Student:	SHRUTI GAUCHANDRA
Roll No. :	15
Assignment No.:	04
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	10

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge Legibility	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : MS. NEHA RAUT

Signature :

Date :

Q1. Display volume of sphere and hemisphere using interface and define the template of methods to be there in the derived class.

→ code:

```
import java.io.*;
import java.util.*;

interface VolumeCalculable {
    double PI = 3.14159;
    double CalculateVolume (double radius);
}

Class Sphere implements VolumeCalculable
{
    @Override
    public double calculateVolume (double radius)
    {
        return (4.0/3.0)* PI* Math.pow (radius, 3);
    }
}

Class Hemisphere implements VolumeCalculable
{
    @Override
    public double calculateVolume (double radius)
    {
        return (2.0/3.0) *PI * Math.pow (radius, 3);
    }
}
```

FOR EDUCATIONAL USE

```
public class Main
{
public static void main (String args[])
{
VolumeCalculable sphere = new Sphere ();
VolumeCalculable hemisphere = new Hemisphere ();
double radius = 5.0;
double sphereVolume = sphere.calculateVolume
                                    (radius);

double hemisphereVolume = hemisphere.calculate
                                    Volume(radius);
System.out.println("Volume of the sphere with
          radius" + radius + "is:" + sphereVolume);

System.out.println("Volume of the Hemisphere
          with radius" + radius + " ~~them~~
                    + hemisphere.Volume);
}
}
```

Output:

Volume of the sphere with radius 5.0 is : 523.598334

Volume of the Hemisphere with radius 5.0 is 261.799167

Q2. Create rectangle and cube class that encapsulates the properties of a rectangle and cube i.e. rectangle has default and parameterized constructor and ~~volume()~~ area() method. Cube has default and parameterized constructor and volume () method. They shared no ancestor other than object. Implement a class size with size () method. This method accepts a single reference agruament z. If z refers to a rectangle then size(z) returns its area and if z is a reference to a cube, then size(z) returns its volume. If z refers to an object of any other class, then size (z) returns -1. Use main() method in size class to call size (.....) method.

→ Code:

```
class Rectangle
{
    private double length;
    private double width;
    public Rectangle()
    {
        this.length = length;
        this.width = width;
    }
public ~~double~~ Rectangle (double length, double width)
{
    this.length = length;
    this.width = width;
}
```

```
    public double area()
    {
    return length * width;
    }
}
class Cube
{
   private double side;
   public Cube()
    {
       this.side = 1.0;
    }
   public Cube(double side)
   {
     this.side = side;
   }
   public double Volume()
    {
     return Math.pow(side, 3)
    }
}

class Size
{
  public double size(Object z)
  {
   if(z instanceof Rectangle)
    {
     Rectangle rect = (Rectangle)z;
     return rect.area();
```

FOR EDUCATIONAL USE

```
}
else if (z instance of Cube)
{
  Cube cube = (Cube) z;
  return cube.volume();
}
else
{
  return -1;
}
}
public static void main (String args[])
{
 Retangle rect1 = new Rectangle();
 Rectangle rect2 = new Rectangle(4.0, 5.0);
 Cube cube1 = new Cube();
 Cube cube 2 = new Cube(3.0);
 Size sizeCalculator = new Size();
 System.out.println("Area of default rectangle:"
            + sizeCalculator.size(rect1));

 System.out.println("Area of parametarized.
   rectangle: " + sizeCalculator.size(rect2));

 System.out.println("Volume of default cube:"
            + size.calculator.size(cube1));

 System.out.println("Volume of parameterized
       cube:" + sizeCalculator.size(cube2));
```

```
String testobj = "Test String";
System.out.println("size of other object
                    (string): "
                  +sizeCalculator.size.size
                    (testobj));

  }
}
```

Output:

Area of default rectangle = 1.0
Area of parameterized rectangle = 20.0
~~Area~~ .
Volume of default cube : 1.0
Volume of parameterized cube : 27.0
Size of other object (string) : -1.0 .